

WORKNEST

Freelance Platform — System Schema Design & ERD Documentation

Document Version: 1.0.1

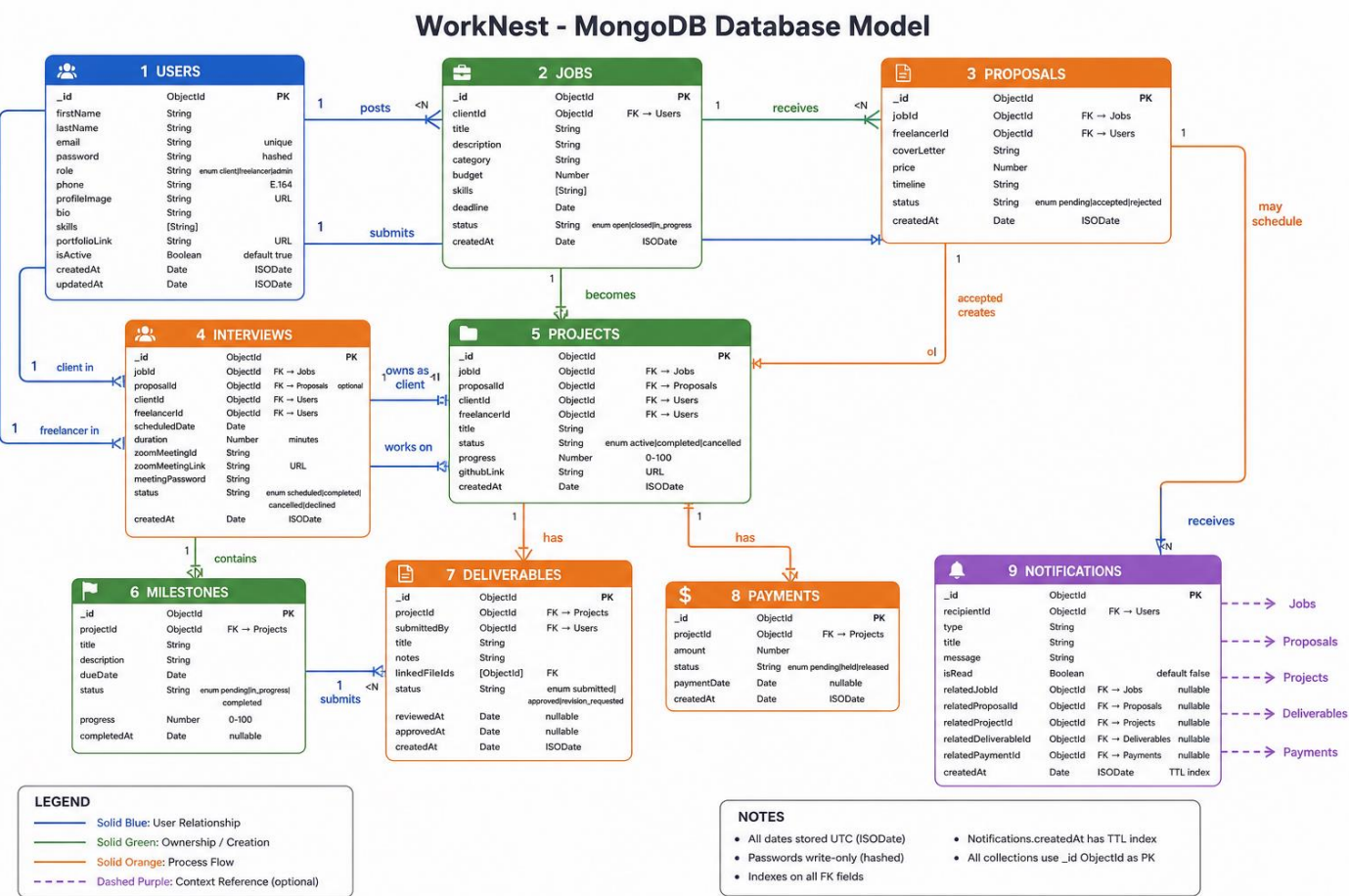
Architecture Type: Document-Oriented (MongoDB)

Date: July 1, 2026

1. Executive Summary & Architectural Paradigm

WorkNest is a high-performance freelance marketplace platform. This document encapsulates the complete relational and structural design of the data layer. Although implemented using a document-oriented database paradigm (MongoDB), the architecture enforces robust relational references, strict status lifecycles, and role-based access controls (RBAC) to ensure absolute transactional integrity across jobs, proposals, contracts, milestones, and payments.

2. Entity-Relationship Diagram (ERD) Overview



3. Comprehensive Collections Schema & Data Dictionary

3.1 Users Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	Root system identity context.
firstName	String	Required Max: 50 chars	—
lastName	String	Required Max: 50 chars	—
email	String	Required Unique Lowercase Trimmed	Primary security/auth identity.
password	String	Required Hashed Min: 8 chars	Write-only / Authentication layer.
role	String	Required Enum: client, freelancer, admin	Drives system RBAC middleware.
phone	String	Optional	E.164 compliance recommended.
profileImage	String	Optional	URL to secure storage bucket.
bio	String	Optional Max: 500 chars	Professional summary introduction.
skills[]	Array (String)	Optional	Indexed vector for algorithmic job matching.
portfolioLink	String	Optional	External developer/portfolio URI.
isActive	Boolean	Default: true	Soft-delete status switch.
createdAt	Date	Auto-generated	—
updatedAt	Date	Auto-generated	—

3.2 Jobs Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	—
clientId	ObjectId	Required	Foreign Key → Users
title	String	Required Max: 150 chars	—
description	String	Required Max: 5000 chars	Markdown formatting allowed.
category	String	Required	System taxonomy classification.
budget	Number	Required Min: 1	USD gross allocation.
skills[]	Array (String)	Required	Required operational proficiencies.
deadline	Date	Required Must be future date	Validation enforced at API gateway.
status	String	Enum: open, closed, in_progress	Workflow engine driver state.
createdAt	Date	Auto-generated	—

3.3 Proposals Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	—
jobId	ObjectId	Required	Foreign Key → Jobs
freelancerId	ObjectId	Required	Foreign Key → Users
coverLetter	String	Required Max: 3000 chars	—
price	Number	Required Min: 1	Freelancer gross proposed value.
timeline	String	Required	Estimated milestone execution duration.
status	String	Enum: pending, accepted, rejected	State modifications update pipeline.
createdAt	Date	Auto-generated	—

3.4 Interviews Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	—
jobId	ObjectId	Required	Foreign Key → Jobs
proposalId	ObjectId	Required	Foreign Key → Proposals
clientId	ObjectId	Required	Foreign Key → Users (Interviewer)
freelancerId	ObjectId	Required	Foreign Key → Users (Interviewee)
scheduledDate	Date	Required	UTC ISO timestamp.
duration	Number	Required	Configured span in minutes.
zoomMeetingId	String	Required	Third-party integrated meeting hash.
zoomMeetingLink	String	Required	Secure integration webhook target URL.
meetingPassword	String	Optional	Plain/token credential.
status	String	Enum: scheduled, completed, cancelled, declined	Workflow transition index.
createdAt	Date	Auto-generated	—

3.5 Projects Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	—
jobId	ObjectId	Required	Foreign Key → Jobs
clientId	ObjectId	Required	Foreign Key → Users
freelancerId	ObjectId	Required	Foreign Key → Users
title	String	Required	Inherited or explicitly overwritten title.
status	String	Enum: active, completed, cancelled	Platform contract tracker state.
progress	Number	Default 0 Range: 0–100	Aggregate metric updated by milestones.
githubLink	String	Optional	Linked VCS repository hook.
createdAt	Date	Auto-generated	—

3.6 Milestones Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	—
projectId	ObjectId	Required	Foreign Key → Projects
title	String	Required Max: 150 chars	—
description	String	Optional	Detailed deliverables criteria scope.
dueDate	Date	Required	Task target time horizon.
status	String	Enum: pending, in_progress, completed	Functional state check flag.
progress	Number	Default 0 Range: 0–100	Individual milestone step completeness.
completedAt	Date	Optional	Populated dynamically on state transition.

3.7 Deliverables Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	—
projectId	ObjectId	Required	Foreign Key → Projects
submittedBy	ObjectId	Required	Foreign Key → Users (Freelancer)
title	String	Required	Brief identifier of work module.
notes	String	Optional Max: 2000 chars	Freelancer delivery context/notes.
linkedFileIds[]	Array (ObjectId)	Optional	Foreign Key → ProjectFiles
status	String	Enum: submitted, approved, revision_requested	Controls review SLA hooks.
reviewedAt	Date	Optional	—
approvedAt	Date	Optional	Triggers automated payment workflows.
createdAt	Date	Auto-generated	Submission timestamp marker.

3.8 Payments Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	—
projectId	ObjectId	Required	Foreign Key → Projects
amount	Number	Required Min: 1	Gross transactional funds amount.
status	String	Enum: pending, held, released	Operational management stage.
paymentDate	Date	Required	Execution ledger timestamp.
createdAt	Date	Auto-generated	System ingestion confirmation stamp.

3.9 Notifications Collection

Field Name	Data Type	Constraints & Validations	Relationships & Notes
_id	ObjectId	Primary Key	—
recipientId	ObjectId	Required	Foreign Key → Users
type	String	Required	Event descriptor (e.g., proposal_received).
title	String	Required	User interface header string.
message	String	Required	Detailed push/alert description content.
isRead	Boolean	Default: false	Read tracking switch.
relatedJobId	ObjectId	Optional	Foreign Key → Jobs (Context)
relatedProposalId	ObjectId	Optional	Foreign Key → Proposals (Context)
relatedProjectId	ObjectId	Optional	Foreign Key → Projects (Context)
relatedDeliverableId	ObjectId	Optional	Foreign Key → Deliverables (Context)
relatedPaymentId	ObjectId	Optional	Foreign Key → Payments (Context)
createdAt	Date	Auto-generated	TTL Index applied standard 30-day prune.

4. Integrity, Cascading Policies & Performance Strategy

To ensure optimum database traversal speeds and clean system operations, the following indexing schema and database governance strategies are mapped across the system architecture:

- **Compound Indexing:** Compound indices must be deployed on { status: 1, createdAt: -1 } across the Jobs, Proposals, and Deliverables collections to resolve active interface queries concurrently.
- **Referential Integrity Hooks:** While native cascading deletes are absent in MongoDB, server-side pre-remove hooks inside the application middleware must strictly intercept User deletions to mark isActive: false, avoiding broken multi-point references.
- **Security Validation:** Complete schema constraints are mapped directly to Mongoose validator models, guaranteeing string boundaries are validated prior to operational layer commit operations.